



# Precipitation and Filtration of lead in Flint Michigan's Water Contamination

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## Introduction

The water crisis in Flint, MI started in April 2014 when a switch was made from treated water through the Detroit Water and Sewerage Department (sourced from Lake Huron and the Detroit River) to the Flint River. Failing to treat this new source of water resulted in a number of public health issues, the most severe being lead contamination. Between 6-12 thousand children have been exposed to contaminated water, increasing the percentage of children with elevated blood-lead levels from 2.5% in 2013 to 5% in 2015.

The human body contains approximately 120mg of lead. About 10-20% of lead is absorbed by the intestines. If overexposed, some of the symptoms are colics, skin pigmentation and paralysis. Generally, effects of lead poisoning are neurological of teratogenic. The highest levels of lead recorded in Flint was 13,000 ppb (parts per billion) in 2015. This level is approximately more than 866 times the federal guideline of 15ppb. Quite recently, Gov. Snyder has announded the end of the free bottled water program in Flint despite the fact that certain neighborhoods in Flint do not have safe drinking water.

## Purpose

The purpose of this research is to find a way how to resolve the water contamination problem in Flint, Michigan. One of the methods used was lead precipitation using KDF85 (Kinetic Degradation Fluxion), a process media that applies high-purity copper zinc granules to reduce contaminants in water by using an oxidation/reduction reaction.

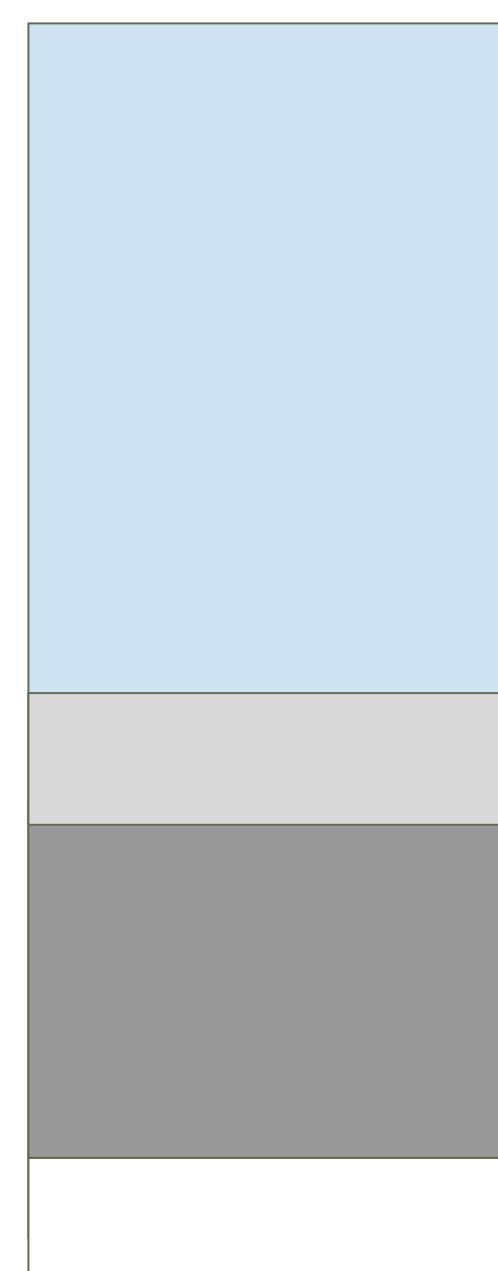
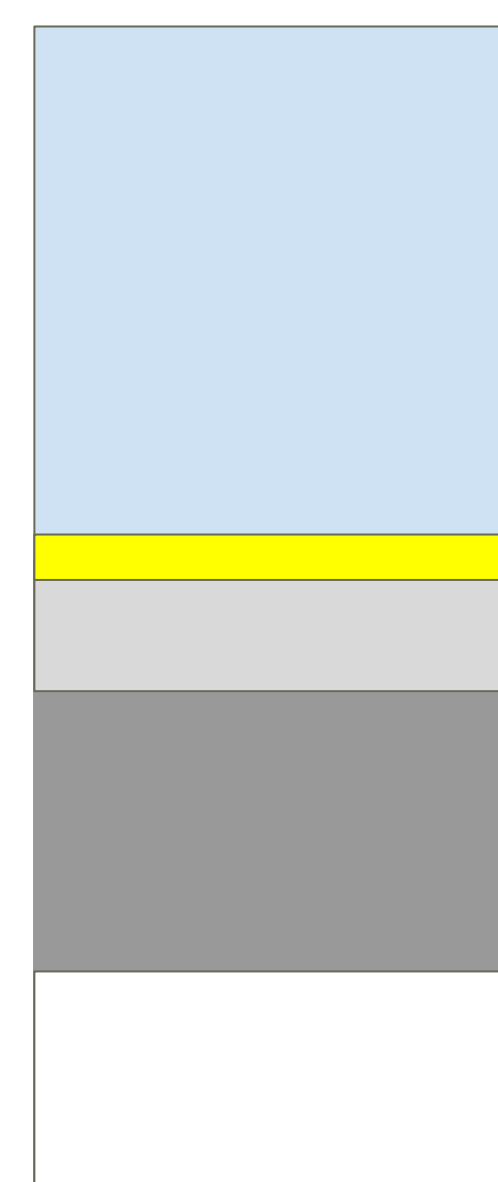
Sodium sulfide is also an excellent reagent in precipitation of lead. Combination of lead sulfide precipitation and KDF 85 media will reduce backwashing KDF and increase efficiency of the system. Zeolite Ferrierite also contributes to the apparatus as one of the most efficient zeolite in the removal of lead (Pb) and other metals.

## Methods

A solution (1) contain lead (II) nitrate with 158 ppm concentration of  $Pb^{2+}$  was made. Two trials were carried out:

(1) A filter-like apparatus is made with 10.00 g zeolite ferrierite, 82.00 g of KDF 85 and 2.025g  $Na_2S$  as shown in figure 2. 50 mL of solution (1) is tested on the apparatus.

**Legend**  
Yellow -  $Na_2S$   
Gray - Zeolite Ferrierite  
Dark Gray - KDF 85  
Blue - Lead Contaminated  $H_2O$



(2) The same structure apparatus is made with 3.00 g zeolite ferrierite and 82.00 g of KDF 85 as shown in figure 2. 100 mL of solution (1) is tested on the apparatus.



Fig.1 - Lead precipitation apparatus

## Results and Discussion

There is a good reduction of lead when treating contaminated water with the apparatuses. The amount of lead is reduced into half using apparatus (1) and to one ninth using apparatus (2). Since the original contaminated solution has 158 ppm lead concentration, treated water from apparatus (1) has 74.35 ppm lead concentration and water from apparatus (1) has 18.59 ppm lead concentration.

In the first trial,  $Na_2S$  reacted with water forming the specific smell of rotten egg. Also the amount of zeolite ferrierite is about three times compared to apparatus (2), which made the received water cloudy because of leftover zeolite powder that could not be filtered completely by cloth and filter papers. Therefore, the amount of zeolite ferrierite was reduced and  $Na_2S$  was not used in the apparatus (2).

| Number of trial | Sample        | Pb | Na      | Mg  | K   | Ca    | Cu |
|-----------------|---------------|----|---------|-----|-----|-------|----|
| 3               | Solution (1)  | 17 | 14.175  | 72  | 66  | 20    | 10 |
| 3               | Apparatus (1) | 8  | 183.230 | 170 | 259 | 1.048 | 28 |
| 3               | Apparatus (2) | 2  | 14.674  | 69  | 70  | 321   | 14 |

Table 1: Testing result from the ICP spectrometer

Including  $Na_2S$  may work best when treated with lead contaminated water alone but is shown not effective in this apparatus. Reducing the amount of zeolite powder surprisingly may have positive or no effect on lead precipitation. Therefore, KDF may has played the biggest role in lead precipitation as the result shown.

Even though the lead was reduced significantly, the amount of other elements were also changed. The amount of calcium from apparatus (2) raised 16 times compared to solution (1), which can cause hard water effect. In solution (1), the amount of magnesium, potassium, sodium and copper increased dramatically. Moreover, the treated water from both apparatus was cloudy, which requires a better post-filter system to remove the leftover zeolite from the water.

## Conclusion

Lead filter apparatus demo was made and yielded positive results. Lead was significantly precipitated out of water mainly by KDF 85, also zeolite ferrierite and  $Na_2S$ . Based on this, more complex and effective apparatus can be built to utilize the use of each component and further purify contaminated water to make drinkable water.

In recent studies, scientist explored the possibility of using wheat straw modified with sodium bicarbonate ( $NaHCO_3$ ) to remove arsenate from an aqueous solution. We look forward to modifying the filtration system in reverse osmosis with sodium bicarbonate in hopes of removing lead contaminants from water. In the future, we are aiming combine different lead treatment methods to eliminate lead out of contaminated water such as the one in Flint Michigan.

## Work Cited

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